

Anatomy of the selected-window border overlap

PRIME.RDAGGER.OVERLAP_MECHANISM.01 — round 467

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Abstract

Round 460 reduced the finite selected-window race to

$$q^\dagger = \sum_j \frac{|\langle u, e_j \rangle|^2}{1 - \lambda_j}.$$

This note tests the proposed smooth-versus-rough mechanism on the production-faithful windows $k = 5, \dots, 12$. The qualitative alignment effect is real: the top MAIN eigenvectors are low-frequency, their small overlaps correlate with lower variation, the ARCH control is almost perfectly aligned, and a deterministic scramble destroys the tiny MAIN alignment. The proposed Abel proof nevertheless fails its own numerical gate. Even the *exact measured* partial sums of u give a three-mode bound 91.703 to 119482.632 times the measured band contribution; the prescribed factor- 10^3 stop gate fires for $k = 7, \dots, 12$.

There is also a decisive type obstruction. The vector u in the spectral formula is the smooth-border functional after the MAIN-dependent orthogonal-polynomial basis change. It is not the raw tent-folded von-Mangoldt lag. Rosser–Schoenfeld controls the source before that nonlinear construction, whereas Abel needs coordinate partial sums after it. No source-to-basis estimate is known. The verdict is therefore

STUCK: QUALITATIVE_ALIGNMENT_REAL;
ABEL_CURRENCY_TOO_COARSE;
SOURCE_TO_BORDER_BASIS_BRIDGE_MISSING.

This is finite research documentation. It does not prove a cofinal overlap bound, the frequently-selected statement, or the Riemann hypothesis. No RH claim and no anti-RH claim is made.

1 Object and sealed diagnostic band

For the production-faithful selected window let B be the negative-channel sampling matrix in the positive-measure orthonormal-polynomial basis and let b be the smooth-border readout. Round 460 uses

$$C = B^T B, \quad u = \frac{b}{\sqrt{B_w}}, \quad C e_j = \lambda_j e_j, \quad \|e_j\|_2 = 1.$$

When $C < I$, spectral calculus gives the displayed formula for $q^\dagger$. The arithmetic source enters both C and u through the construction of the orthonormal basis. In particular, there is no coordinate identity $u = c^P$.

The diagnostic edge band is sealed as the three largest eigenvalues. Equivalently, for each k it is the adaptive interval $\lambda > 1 - \delta_k$, where δ_k is the gap of the third-largest mode. This avoids choosing a threshold after seeing the overlaps. At the two smallest dimensions the third mode is not close to one; it is retained as a conservative control and not advertised as an asymptotic edge mode.

For a unit vector $e = (e_0, \dots, e_{N-1})$ define

$$\text{TV}(e) = \sum_{r=0}^{N-2} |e_{r+1} - e_r|.$$

The reported low-frequency share is the squared real-FFT mass in the lowest quarter of available frequencies. The contribution and budget share are

$$q_j = \frac{|\langle u, e_j \rangle|^2}{1 - \lambda_j}, \quad s_j = q_j / q^\dagger.$$

2 Anatomy

Table 1: Top-three MAIN modes. Rank 1 is the largest eigenvalue.

k	rank	λ	TV(e)	low-F	$ \langle u, e \rangle ^2$	q_j	s_j
5	3	0.000279966	3.568133	0.124051	$1.982 \cdot 10^{-1}$	0.198237	0.225809
5	2	0.388856149	1.955740	0.802224	$7.794 \cdot 10^{-3}$	0.012753	0.014527
5	1	0.826952501	1.270120	0.962359	$7.953 \cdot 10^{-3}$	0.045956	0.052348
6	3	0.312184237	2.597238	0.192129	$6.818 \cdot 10^{-2}$	0.099123	0.111958
6	2	0.680038222	1.824741	0.851250	$1.047 \cdot 10^{-2}$	0.032714	0.036950
6	1	0.944661181	1.216460	0.956185	$1.745 \cdot 10^{-6}$	0.000032	0.000036
7	3	0.601795264	2.325423	0.162044	$9.414 \cdot 10^{-4}$	0.002364	0.002748
7	2	0.823487691	1.747610	0.842646	$6.219 \cdot 10^{-4}$	0.003523	0.004095
7	1	0.986783879	1.207106	0.949658	$5.549 \cdot 10^{-4}$	0.041989	0.048797
8	3	0.825919862	2.341688	0.081557	$2.638 \cdot 10^{-4}$	0.001515	0.001778
8	2	0.915208145	1.455195	0.898819	$1.285 \cdot 10^{-3}$	0.015151	0.017780
8	1	0.994176300	1.239744	0.925906	$2.756 \cdot 10^{-4}$	0.047331	0.055543
9	3	0.973679227	1.878080	0.975438	$3.711 \cdot 10^{-5}$	0.001410	0.001571
9	2	0.990153274	1.175584	0.991145	$2.009 \cdot 10^{-5}$	0.002041	0.002274
9	1	0.999132787	1.013267	0.981384	$1.217 \cdot 10^{-5}$	0.014032	0.015633
10	3	0.978303778	1.910413	0.966895	$3.450 \cdot 10^{-5}$	0.001590	0.001706
10	2	0.996633709	1.276908	0.990047	$4.624 \cdot 10^{-5}$	0.013735	0.014738
10	1	0.999586415	1.078482	0.974490	$2.736 \cdot 10^{-6}$	0.006614	0.007097
11	3	0.987897150	1.527471	0.994031	$2.957 \cdot 10^{-7}$	0.000024	0.000027
11	2	0.998679332	1.270301	0.985855	$7.412 \cdot 10^{-6}$	0.005612	0.006099
11	1	0.999826747	0.941445	0.982699	$2.437 \cdot 10^{-6}$	0.014066	0.015286
12	3	0.995768405	1.454589	0.997738	$2.786 \cdot 10^{-6}$	0.000658	0.000720
12	2	0.999290213	1.224994	0.988744	$8.486 \cdot 10^{-6}$	0.011956	0.013082
12	1	0.999912794	0.887150	0.988762	$6.725 \cdot 10^{-7}$	0.007712	0.008438

The top-mode low-frequency share is 0.926–0.989. Thus the first anatomical premise is true on this finite census. However, the fitted variation law is

$$\text{TV}(e_{\text{top}}) \asymp N^{-0.196225},$$

not $O(N^{-1})$. Nor would an $O(N^{-1})$ theorem follow merely from “finite Jacobi edge mode”: $C = B^T B$ is generally dense in this basis, not the Jacobi multiplication matrix. Even for tridiagonal Jacobi matrices, coefficient regularity and delocalization hypotheses would be needed to exclude localized edge eigenvectors.

The correlation between $\text{TV}(e)$ and $\log_{10} |\langle u, e \rangle|^2$ is +0.745344 on the eight top modes and +0.702626 on all 24 tabulated modes. This supports an association, not a causal inequality or an asymptotic law.

3 Controls

ARCH. The smooth ARCH control aligns to its top mode at 0.995813–0.998970, exactly in the hypothesized direction. Its normalized u profile is also smoother than MAIN. At full depth the ARCH contraction has eigenvalues above one and its signed chain is not a positive comparison object, as already recorded in round 460. Only alignment, not an ARCH $q^\dagger$ budget, is compared here.

SCRAMBLE. The deterministic source-position scramble uses seed $46701 + k$, preserves the channel weights and border atoms, and changes the arithmetic positions before rebuilding the

Table 2: Normalized top overlap $|\langle u, e_{\text{top}} \rangle|^2 / \|u\|_2^2$ and normalized variation $\text{TV}(u) / \|u\|_2$.

k	MAIN	ARCH	SCRAMBLE	$\text{TV}(u_M) / \ u_M\ $	$\text{TV}(u_A) / \ u_A\ $	$\text{TV}(u_S) / \ u_S\ $
5	$9.525 \cdot 10^{-3}$	0.998055	0.298492	3.705	0.960	3.596
6	$2.216 \cdot 10^{-6}$	0.998236	0.229650	3.906	0.939	4.812
7	$8.895 \cdot 10^{-4}$	0.997602	0.025356	4.315	0.933	4.900
8	$5.050 \cdot 10^{-4}$	0.998970	0.052190	4.524	0.959	2.491
9	$1.936 \cdot 10^{-5}$	0.998756	0.321934	7.072	0.845	3.604
10	$4.152 \cdot 10^{-6}$	0.997964	0.659157	7.777	0.837	4.170
11	$5.104 \cdot 10^{-6}$	0.998159	0.023869	7.606	0.852	3.829
12	$1.545 \cdot 10^{-6}$	0.995813	0.818070	7.533	0.838	4.191

orthogonal basis. Its normalized top overlap is 28.5 to $5.3 \cdot 10^5$ times MAIN. This proves that the tiny MAIN alignment is not a generic consequence of dimension or normalization. It does *not* isolate roughness: the scrambled top-eigenvector variation is not held fixed, and the scrambled u is often rough while its overlap is much larger. The control therefore supports arithmetic specificity but refutes a monotone slogan “rougher u means smaller overlap” as a sufficient mechanism.

4 The exact Abel inequality

Lemma 1 (Finite summation by parts). *For $u, e \in \mathbb{R}^N$, let*

$$U_r = \sum_{i=0}^r u_i, \quad 0 \leq r < N.$$

Then

$$\langle u, e \rangle = U_{N-1} e_{N-1} + \sum_{r=0}^{N-2} U_r (e_r - e_{r+1}),$$

and consequently

$$|\langle u, e \rangle| \leq \max_{0 \leq r < N} |U_r| (|e_{N-1}| + \text{TV}(e)).$$

Proof. Write $u_r = U_r - U_{r-1}$ with $U_{-1} = 0$, expand the scalar product, and shift the second finite sum. The inequality is the triangle inequality applied to the resulting identity. $\square$

Corollary 1 (Finite edge-band bound). *For any set J of modes with $\lambda_j < 1$, put $A = \max_r |U_r|$ and $V_j = |e_{j,N-1}| + \text{TV}(e_j)$. Then*

$$\sum_{j \in J} \frac{|\langle u, e_j \rangle|^2}{1 - \lambda_j} \leq A^2 \sum_{j \in J} \frac{V_j^2}{1 - \lambda_j}.$$

Proof. Square Lemma 1, divide by the positive spectral gaps, and sum. $\square$

This corollary is exact but is not the requested source lemma: A is computed after the MAIN basis change. It is nevertheless the strongest possible test of whether the proposed Abel currency is numerically close, because replacing the exact A by a classical source majorant cannot repair a bound that already fails with the actual coordinate partial sums.

5 Sharpness and stop gate

The classical input itself is available. Rosser and Schoenfeld give

$$\psi(x) \leq c_{\text{RS}} x, \quad c_{\text{RS}} = 1.03883.$$

Table 3: Abel chain. “ ψ factor” optimistically assigns transfer constant one to the Rosser–Schoenfeld source bound; it is a diagnostic, not a valid theorem.

k	top Abel/actual	top ψ /actual	measured band	Abel band	bound/actual
5	13.463	188.357	0.256946	23.563	91.703
6	794.324	16895.197	0.131868	35.437	268.726
7	36.084	1283.954	0.047877	71.763	1498.908
8	42.137	2508.412	0.063997	107.264	1676.071
9	156.513	13576.132	0.017482	427.972	24480.315
10	337.305	43188.277	0.021940	932.552	42505.430
11	285.312	56650.908	0.019702	1469.714	74595.322
12	492.912	143615.580	0.020326	2428.609	119482.632

The finite sieve verifies $\max_{5 \leq k \leq 12} \psi(2^k)/2^k = 1.018864 < c_{\text{RS}}$. Partial summation also gives

$$\sum_{n \leq x} \frac{\Lambda(n)}{\sqrt{n}} \leq c_{\text{RS}}(2\sqrt{x} - 1).$$

What is missing is an inequality transporting this raw-source estimate through lag folding, signed moment formation, positive-channel orthogonalization, and the border readout to $\max_r |\sum_{i \leq r} u_i|$. Calling that transport constant one already produces the bad third column of Table 3; no such constant-one theorem is asserted.

The user’s decision rule was: continue if the bound is within factor 10, and stop honestly if it is worse than factor 10^3 . No top-mode factor is below 10. The full three-mode Abel factor exceeds 10^3 on each $k = 7, \dots, 12$ and reaches 119482.632. The route therefore stops. No lemma of the requested form

$$\sum_{\lambda_j > 1-\delta} \frac{|\langle u, e_j \rangle|^2}{1 - \lambda_j} \leq B_{\text{safe}}$$

with source-explicit constants has been reached.

6 Conditioning, determinism, and alias audit

All eight decompositions were repeated by symmetric eigensolution at 80 decimal digits after exact conversion of the binary64 matrix entries. Across the top-three modes, the maximum eigenvalue discrepancy is $7.288 \cdot 10^{-16}$, the maximum overlap-square discrepancy is $2.119 \cdot 10^{-14}$, and the maximum binary64 symmetric residual is $1.310 \cdot 10^{-15}$. These are diagnostics, not interval enclosures of the source construction; no proof conclusion depends on them. The interval-certified finite $q^\dagger < 1$ statements remain those of round 465.

The probe has fixed seeds, fixed top-mode count, fixed FFT cutoff, sealed pins, and no zero data. Two runs are required to produce byte-identical numerical lines.

Audit against the thirty-one closed routes. The new *anatomy* is not the old unsigned $|\mathcal{Z}_{\text{loc}}|$ triangle: it measures the spectral projector coordinates that round 460 identified. However, the proposed proof step is not wholly new. The r429 catalog already contains “Chebyshev ψ / left–right Abel” and r460 contains raw-resolvent majorization. Once $\max |U_r|$ is replaced by a source majorant and the mode sum is bounded termwise, the present route enters exactly that triangle/majorant class. The claimed novelty survives only as the empirically observed test-direction alignment; it does not survive as a source-derived proof.

The remaining twenty-eight routes in the catalog use different coordinates or quantifiers: fold pairing, finite-head Chebyshev, Gershgorin/Schur assist, OSC–Geronimus, checkerboard, gap masks, source Weyl energy, FORM_T/FORM_P, moment/Euler Gram, rank-one Loewner, Nyquist density, Krein transport, Fourier/Bernstein test polynomials, saturation, hole modes, de Branges phase,

denominator gaps, vacuous-chart coordinates, and finite fixed- r blocks. None is used as a hidden premise here. In particular there is no PNT replacement density, Potapov product, octave additivity, or inferred cofinal monotonicity.

7 Kill table and exact remaining gap

Gate	Result	Reason
Top-mode smoothness	PASS (finite)	low-frequency share 0.926–0.989; TV exponent -0.196225
Overlap–variation correlation	PASS (measured)	$+0.745344$ top, $+0.702626$ all 24
ARCH control	PASS	normalized top alignment 0.995813–0.998970
SCRAMBLE control	PASS with caveat	tiny MAIN alignment destroyed, but roughness and eigenprofile are not independently held fixed
$O(N^{-1})$ variation premise	REFUTED on census	measured exponent -0.196225 ; no applicable Jacobi theorem for dense C
Abel factor below 10	FAIL	best top-mode factor 13.463
Abel stop threshold	FIRED	band factor $> 10^3$ for $k = 7, \dots, 12$
Source-to- u bridge	MISSING	orthogonal-basis transformation is MAIN-dependent and nonlinear
dps conditioning	PASS	80-digit discrepancies and residual below $2.2 \cdot 10^{-14}$
Alias audit	FAILS novelty claim	proof step re-enters r429 Abel/majorant class; anatomy remains new
Determinism	PASS	fixed seed and sealed numerical pins

Remaining gap, verbatim. *Prove, from the tent-folded von-Mangoldt source and with constants uniform on an infinite subsequence of selected windows, a bound on the post-orthogonalization partial sums $\max_r |\sum_{i \leq r} u_{k,i}|$ that simultaneously beats the shrinking spectral gaps; or replace coordinate Abel summation by a cancellation estimate that retains the joint MAIN dependence of u_k and the edge projector. Neither statement is proved here.*

References

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